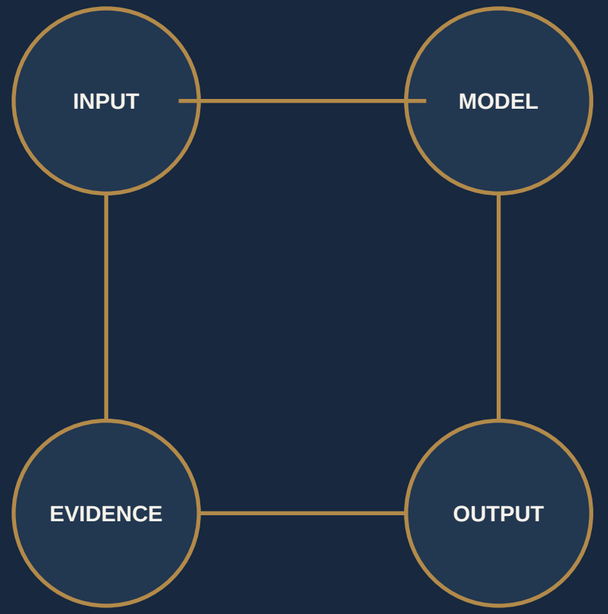
BACHELOR OF ARTIFICIAL INTELLIGENCE & ROBOTICS ENGINEERING

Computer Vision

AI.104 | University lecture series



What you will learn

A practical progression from core ideas to engineering judgment

- Explain and define image formation.
- Apply key methods including pixels.
- Build or analyze a practical system involving color spaces.
- Evaluate assumptions, limitations, performance and responsible use.

A connected learning journey

Each stage prepares the tools needed for the next.

01

FOUNDATIONS

Image formation

02

METHODS

Pixels

03

SYSTEMS

Color spaces

04

PRACTICE

Laboratory, evaluation and project

The subject at a glance

A map of the ideas and techniques in this course

01 · Image formation

Image formation

02 · Pixels

Pixels

03 · Color spaces

Color spaces

04 · Filtering

Filtering

05 · Edge detection

Edge detection

06 · Feature extraction

Feature extraction

07 · Segmentation

Segmentation

08 · Object detection

Object detection

Core ideas, made explicit

AI.104 • Distinguish definitions, assumptions and evidence

- Image formation
- Pixels
- Color spaces
- Filtering
- Edge detection

DEFINE → MODEL → TEST → INTERPRET

Core ideas, made explicit

AI.104 • Distinguish definitions, assumptions and evidence

- Feature extraction
- Segmentation
- Object detection
- Image classification
- Tracking and introductory deep-learning vision

DEFINE → MODEL → TEST → INTERPRET

Follow the system through its stages

Use the process diagram to connect components and outcomes.

PROBLEM
Image formation

INPUTS
Pixels

METHOD
Color spaces

OUTPUT
Filtering

EVALUATION
Evidence, constraints and limitations

Teaching sequence: reveal one component at a time; then replay the complete end-to-end flow.

From concept to a real engineering decision

A short case study for discussion and worked-example narration

SCENARIO

A real task involving image formation.

CONSTRAINTS

Data, resources, safety and operating conditions shape the design.

DECISION

Select a method; justify the trade-offs and define success.

- State assumptions before calculating or implementing.
- Compare the expected result with observed evidence.

Laboratory · build, measure, improve

Students work through the following guided sequence.

- Specify a measurable task using image formation.
- Implement or simulate a baseline using pixels.
- Run a controlled test and record results with units and conditions.
- Diagnose a failure, revise the design and explain the improvement.

Review · explain · apply

KNOWLEDGE CHECK

What assumptions does the method make? What evidence would show that it works?

PRACTICAL CHALLENGE

Apply one course idea to a new context; state constraints, test criteria and likely failure modes.

Key takeaway: make the model visible, test it, and communicate its limits.

Course-series slide design • detailed narration, demonstrations and assessments expand each module.